

EX:NO:27

PROGRAM:

```
import itertools
```

```
def assignment_problem(cost):
```

```
    n = len(cost)
```

```
    min_cost = float('inf')
```

```
    best = None
```

```
    for perm in itertools.permutations(range(n)):
```

```
        total = sum(cost[i][perm[i]] for i in range(n))
```

```
        if total < min_cost:
```

```
            min_cost = total
```

```
            best = perm
```

```
    return best, min_cost
```

```
# Test
```

```
cost = [[3,10,7],[8,5,12],[4,6,9]]
```

```
print(assignment_problem(cost))
```

Output

```
((2, 1, 0), 16)
```

```
=== Code Execution Successful ===
```